

Suhani Verma

ECE Undergraduate (2023–2027)

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About Me

ECE undergraduate with hands-on experience in AI/ML, signal processing, radar systems, embedded platforms, and ML-driven cybersecurity. Worked on multilingual NLP, intrusion detection systems, CFAR-based radar simulations, and multimodal AI architectures through IIT research and industry internships. Strong foundation in Python, C/C++, MATLAB, and ML frameworks with interest in applied ML, hardware-aware AI, autonomous systems, and defense technologies.

Research & Technical Interests

- Applied Machine Learning for Systems & Security
- Signal Processing, Radar Systems, and CFAR Detection
- Embedded and Hardware-Aware AI
- Multi-Agent and Distributed Systems
- NLP for Low-Resource Languages

Education

B.Tech in Electronics and Communication Engineering	2023–2027 (Exp.)
Banasthali Vidyapith, Rajasthan	CGPA: 7.0/10
AISSCE (Class XII)	2023
DPS Ranipur, Haridwar	Percentage: 82.94%
AISSE (Class X)	2020
DPS Ranipur, Haridwar	Percentage: 94.75%

Research Publications

- **Multi-Agent MBTI-Inspired Cognitive AI for Cybersecurity Threat Detection.** SSRN (2025)
- **Economic Implications of Indigenous Weapon Development: DRDO Case Study.** SSRN (2025)
- **GraMLID: GRU-Assisted Multilingual BERT for Word-Level Language Identification in Low-Resource Dravidian Texts.** FIRE (2025)

Internship & Research Experience

IIT Mandi — Computer Vision Research Intern	2026
• Worked on RGB–LiDAR perception pipelines and multimodal scene understanding. • Built training, inference, and visualization pipelines using PyTorch and OpenCV.	
IIT Tirupati — Antenna Design Research Intern	2026
• Designed and analyzed RF antenna architectures for wireless communication systems. • Performed simulation and electromagnetic analysis using MATLAB tools.	
IIT BHU — AI/NLP Research Intern	2025
• Developed multilingual NLP pipelines for low-resource Dravidian text processing. • Evaluated transformer-based models using precision, recall, and F1-score metrics.	
S.K. Infotech	2024
• Developed Java Swing GUI applications using MVC architecture and debugging workflows.	
IBM — AI in Cybersecurity Intern	2022
• Studied anomaly detection models and structured feature pipelines for intrusion detection systems.	

Key Projects

Bidirectional RGB–LiDAR Translation System (under Publication)	<i>PyTorch, OpenCV</i>
Designed pipelines for bidirectional RGB to LiDAR translation for autonomous perception and sensor fusion.	
Neuromorphic AI-Powered Prosthetic Hand (under Publication)	<i>ESP32, Embedded AI</i>
Developed a GRU-based prosthetic control system using EMG, flex, and FSR sensor signals.	
AI-Powered Swarm Drone System (VERONICA)	<i>Python, Simulation</i>
Built a swarm drone simulation with radar tracking, communication, and autonomous coordination.	
Radiology and Gastrointestinal AI System (under Publication)	<i>Transformers, CV</i>
Developed multimodal medical AI pipelines integrating computer vision and transformer-based architectures.	
MBTI-Inspired Multi-Agent Cognitive AI (Published)	<i>Python, ML</i>
Built a multi-agent IDS achieving 90% accuracy and 0.902 F1-score on UNSW-NB15 using RandomForest-based pipelines.	
Adaptive Radar Signal Detection using CFAR	<i>MATLAB, Radar</i>
Simulated adaptive radar systems implementing CFAR-based thresholding for dynamic target detection.	
GraMLID: GRU-Assisted Multilingual BERT for Language Identification (Published)	<i>PyTorch, NLP</i>
Developed a GRU-enhanced multilingual BERT framework for low-resource Dravidian language identification under FIRE 2025.	
Wireless 5G Communication System	<i>IoT, MQTT, YOLO</i>
Developed low-latency smartphone communication and real-time activity visualization systems.	
Antenna Design and RF Simulation (under Publication)	<i>RF, MATLAB, HFSS</i>
Designed and analyzed antenna architectures emphasizing RF performance and EM behavior.	
MotionAI: Human Activity Recognition	<i>LSTM, CNN</i>
Implemented deep learning pipelines for time-series human activity classification.	
Invio: AI Accessibility Assistant	<i>Computer Vision</i>
Developed an AI-powered accessibility solution for visually impaired users using intelligent sensing.	
FRIDAY: Intelligent NLP Voice Assistant	<i>Python, NLP</i>
Built a conversational AI assistant using NLP, speech recognition, and automation workflows.	
JARVIS AI Automation System	<i>Python, Automation</i>
Engineered an AI-driven virtual assistant integrating NLP and adaptive task execution.	
Tense Sense: NLP-Based Tense Classification System	<i>ML, NLP</i>
Designed a machine learning model for automatic sentence tense classification.	
Database Inventory Management System	<i>Python, SQL</i>
Developed a SQL-integrated inventory management platform for sales and database operations.	

Technical Skills

Programming	Python, C, C++, Java, SQL, MATLAB
ML/AI	TensorFlow, PyTorch, scikit-learn, Transformers, OpenCV
Signal Processing	CFAR Detection, Signal Preprocessing, MATLAB Radar Toolbox
Embedded Systems	Arduino, Raspberry Pi
Core Concepts	Data Structures & Algorithms, OOP, MVC, Debugging, Performance Optimization

Certifications & Leadership

Certifications	Leadership & Achievements
• AWS Academy Cloud Foundations	• Silver Medalist — Basketball
• MATLAB Certified Course	• IT Head, DPS Ranipur — IT Council Award
• DreamBuilder Entrepreneurship (2x)	• Open-Source Contributor — GSSoC, Hacktoberfest

References

Dr. Sukomal Pal — Associate Professor, IIT BHU
Dr. Supriya Chanda — Assistant Professor, Bennett University
Dr. Dinesh Singh - Assistant Professor, IIT Mandi
Dr. Ramesh Singh - Assistant Professor, IIT Tirupati